

The logo for Oracle Academy is centered on a light gray background. It features the word "ORACLE" in a bold, orange, sans-serif font. Below it, the word "Academy" is written in a smaller, dark gray, sans-serif font. The entire logo is framed by a thin black border, with dark gray horizontal bars at the top and bottom.

ORACLE

Academy

Database Foundations

2-4

Unique Identifiers

ORACLE
Academy



Copyright © 2020, Oracle and/or its affiliates. All rights reserved.

Objectives

- This lesson covers the following objectives:
 - Identify unique identifiers (UIDs)
 - Identify artificial unique identifiers
 - Identify composite unique identifiers
 - Identify candidate and secondary unique identifiers
 - Define primary keys

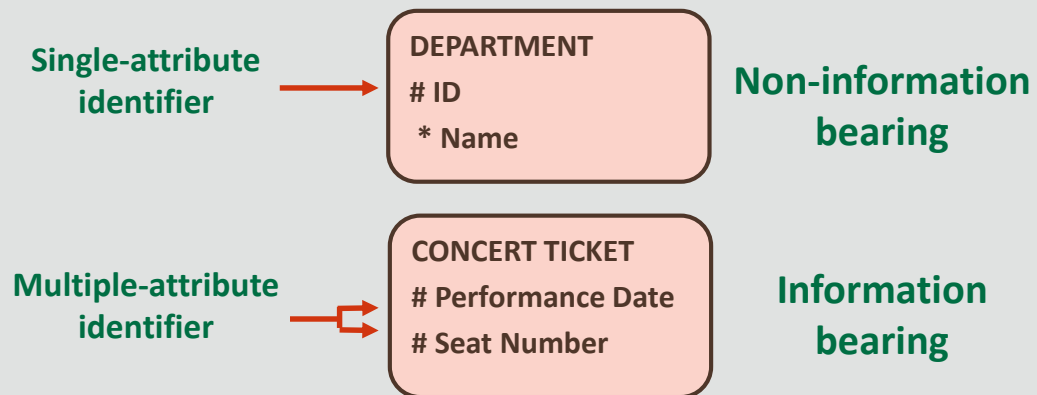


Unique Identifiers

- A unique identifier is an attribute of an entity that meets the following rules:
 - It is unique across all instances of the entity
 - It has a non-NULL value for each instance of the entity for the lifetime of the instance
 - It has a value that never changes for the lifetime of the instance
- A UID is a special attribute or group of attributes that uniquely identifies a particular instance of an entity

Unique Identifier: Examples

- Each entity must have a unique identifier
- Otherwise, it is not an entity



The UID can be information bearing, when its values have some business meaning. A non-information-bearing UID is sometimes referred to as a artificial key. An information-bearing UID is sometimes referred to as a natural key.

Simple UUIDs vs Composite UUIDs

- A UUID that is a single attribute is a simple UUID
- Sometimes a single attribute is not enough to uniquely identify an instance of an entity
- If the UUID is a combination of attributes its called a composite UUID

Simple UUID



CONCERT TICKET
Ticket Number
* Name

Composite UUID



CONCERT TICKET
Performance Date
Seat Number

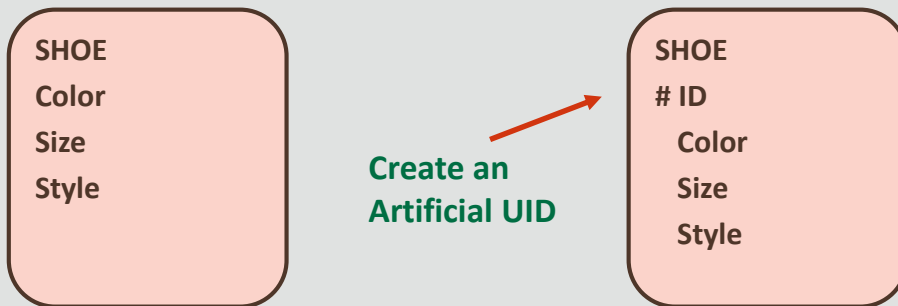
Project Exercise 1

- DFo_2_4_1_Project
 - Oracle Baseball League Store Database
 - Identifying Unique Identifiers (UIDs)



Artificial Unique Identifier

- An artificial UID is made from data that is assigned or generated by the system
- Artificial UIDs do not occur in the natural world but are created for identification purposes in a system



Project Exercise 2

- DFo_2_4_2_Project
 - Oracle Baseball League Store Database
 - Identifying Artificial Unique Identifiers



Case Scenario

MEMBER

First Name

Last Name

Street Address

How can you uniquely identify a member?

Can you use a combination of first name and last name? Only if you are sure that the combination is unique.

MEMBER

First Name

Last Name

Street Address

Case Scenario

MEMBER

ID

First Name

Last Name

Address

First Name and Last Name can be the same for more than one member. So, let's create an artificial UID called ID

Candidate Unique Identifiers

- An entity can have more than one UID
- Candidate UIDs:
 - Badge number
 - Payroll number
- Only one of the candidate UIDs can be chosen as the primary UID
- The other candidates are called secondary UIDs

EMPLOYEE

Badge Number
Payroll Number
First Name
Last Name

There is no standard diagramming convention for tagging secondary UIDs.

Case Scenario

MEMBER

ID

Email

First Name

Last Name

Street Address

ID has been chosen as the primary UID in the Member entity. Can you identify the candidate UID?

Case Scenario

Email can be chosen as the candidate UID because email is unique for every member.

MEMBER

ID
(#) Email
First Name
Last Name
Street Address

Primary Key

- A UID becomes a primary key when the logical model is transformed into a physical database
- A primary key (PK) is a column or set of columns that uniquely identifies each row in a table
- It cannot contain null values
- A PK is either an existing table column or a column that is specifically generated by the database according to a defined sequence
- It must contain a unique value for each row of data



Primary Key: Example

EMPLOYEES

EMPLOYEE_ID	FIRST_NAME	LAST_NAME		DEPARTMENT_ID
100	Steven	King		90
101	Neena	Kochhar		90
201	Rick	Bel		90
205	Shelly	Higgins		10
300	Bill	Steveward		110

↑
Single-Column PK

Composite Primary Key: Example

ACCOUNTS

BANK_NO	ACCT_NO	BALANCE	DATE_OPENED
104	71432	12,000	29-Oct-90
104	34578	18,000	12-Sep-85
105	78967	20,000	
103	96545	60,000	04-Aug-91
105	72345	10,000	03-Jan-99



These fields are combined to form a “composite” PK – a PK consisting of multiple fields

Summary

- In this lesson, you should have learned how to:
 - Identify unique identifiers (UIDs)
 - Identify artificial unique identifiers
 - Identify composite unique identifiers
 - Identify candidate and secondary unique identifiers
 - Define primary keys



The logo for Oracle Academy is centered on a light gray background. It features the word "ORACLE" in a bold, orange, sans-serif font. Below it, the word "Academy" is written in a smaller, dark gray, sans-serif font. The entire logo is framed by a thin black border, with dark gray horizontal bars at the top and bottom.

ORACLE

Academy